

Curriculum Vitae

Zihan (Hazel) Liu

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EDUCATION

- **PhD student in Informatics, University of Illinois Urbana-Champaign** Illinois, US
August 2024 - Present
- **Bachelor's Degree in Software Engineering, Sichuan University** Chengdu, China
GPA: 3.74/4; September 2019 - June 2023

RESEARCH INTERESTS

How conversational AI shapes human meaning-making processes in interpretive domains — contexts where meaning is subjective and without a single correct answer, such as humor, social norms, and emotional uncertainty — and what this implies for how people think, judge, and communicate with and through AI.

PUBLICATIONS

- **Understanding Public Perceptions of AI Conversational Agents: A Cross-Cultural Analysis.**
2024 CHI conference on human factors in computing systems (CHI'24).
Authors: [Zihan Liu](#), Han Li, Anfan Chen, Renwen Zhang, Yi-Chieh Lee
- **The Social Gaze of LLMs: A Literature Review of Multimodal Approaches to Human Behavior Understanding.**
arXiv preprint (2025)
Authors: [Zihan Liu](#), Parisa Rabbani, Veda Duddu, Kyle Fan, Madison Lee, Yun Huang
- **AI-Enhanced Speech-Language Intervention Documentation: Opportunities and Design Goals.**
International Conference on Artificial Intelligence in Education.(2025)
Authors: Qingxiao Zheng, Abhinav Choudhry, [Zihan Liu](#), Parisa Rabbani, Yuting Hu, Abbie Olszewski, Yun Huang, Jinjun Xiong
- **Understanding How Psychological Distance Influences User Preferences in Conversational versus Web Search.**
2025 CHI conference on human factors in computing systems (CHI'25).
Authors: Yitian Yang, Yugin Tan, Yang Chen Lin, Jung-Tai King, [Zihan Liu](#), Yi-Chieh Lee
- **AI Privacy in Context: A Comparative Study of Discourse on Conversational AI Privacy on US and Chinese Social Media.**
The special issue "Comparative Approaches to Studying Privacy: Opening Up New Perspectives" in Social Media + Society. (2024)
Authors: Renwen Zhang, Han Li, Anfan Chen, [Zihan Liu](#), Yi-Chieh Lee
- **A Multi-Objective Genetic Optimization Algorithm for Charger Selection in Static Charger Deployment Scheme for WRSN**
2022 IEEE 14th International Conference on Advanced Infocomm Technology (ICAIT).
Authors: Hong Liu, Lingyun Zhong, [Zihan Liu](#), Feng Lin

RESEARCH EXPERIENCE

- **Research intern advised by Prof. Yi-Chieh (EJ) Lee** National University of Singapore
Interests: Computational Social Science, Conversational Agents Jan 2023 - Jan 2024
 - Led a large-scale computational study of public perceptions of conversational AI, independently handling theoretical grounding, data collection, analysis, and writing throughout (CHI'24, first author). The study applied BERTopic and Word2Vec-based word embedding to analyze nearly one million social media posts from Twitter and Weibo, revealing systematic cross-cultural differences in how users in the US and China attribute warmth, competence, and emotional valence to AI agents — and how these perceptions interact with technical characteristics such as conversational modality, physical embodiment, and degree of anthropomorphism.
 - Also contributed to two related projects during this period. One examined how psychological distance shapes user preferences between conversational and web search interfaces (CHI'25), where I participated in research discussions and contributed to ideation. In another project investigating whether conversational agents with pre-set perspectives influence users' epistemic stance, I led the literature review, co-designed the experimental protocol, and conducted data analysis — including topic modeling of interview transcripts and analysis of survey data on users' self-efficacy and perception of their interlocutor.
- **PhD Research** University of Illinois Urbana-Champaign
Interests: Multimodal LLM, Social Intelligence, Behavior Understanding Aug 2024 - Present
 - Conducted a systematic literature review of LLM-powered multimodal systems for human social behavior understanding, reviewing 176 papers across healthcare, education, HCI, and entertainment (arXiv preprint, first author). Led the project from scope definition through paper writing; introduced a four-dimensional coding framework grounded in psychological theories of social intelligence, examining how systems operationalize competencies such as social perception, reasoning, and interaction. Key findings include a pervasive modality-to-text bottleneck that strips away nuanced social cues, an over-reliance on machine-centric benchmarks, and systematic under-treatment of societal risks such as deception and fairness.

- Also contributed to a study examining how MLLMs can assist speech-language pathologists (SLPs) in documenting and evaluating pediatric therapy sessions (AIED'25). Based on analysis of SLPs' clinical needs, I developed a structured prompt cheat sheet that translates practitioners' assessment logic into prompting strategies, addressing three core challenges defined as Contextual Fluidity, Assessment Fluidity, and Individual Fluidity. The cheat sheet surfaced a deeper finding: accurate AI-assisted behavior interpretation requires embedding the kind of nuanced, person-specific knowledge that clinicians carry — knowledge that resists easy formalization.
- Developing a research program examining how conversational AI reshapes users' cognitive and emotional processes during moments of uncertainty and meaning-making — specifically how AI-mediated interaction alters the sensemaking process itself, and what downstream effects this has on users' reasoning, judgment, and decision-making. This work sits at the intersection of human-AI interaction and cognitive/epistemic wellbeing, with implications for how AI systems should be designed to support rather than substitute for genuine understanding.

PROFESSIONAL EXPERIENCE

- **IKE Environment Technology Co., Ltd. - Intern** Chengdu, China
Front-end Development *January 2021 - February 2021*
 - Contributed to front-end development of web applications, acquiring self-taught proficiency in HTML, CSS, JavaScript, and Vue. Developed a foundational understanding of the application development lifecycle, encompassing requirement analysis, prototyping, and coding.

TEACHING & SERVICE

- **External reviewer** ACM CHI Conference on Human Factors in Computing Systems (CHI 2026)
- **Graduate Teaching Assistant** IS 457: Data Storytelling (Fall 2025 & Spring 2026)

SKILLS

- **Programming & Frameworks** Python, Java, C, SQL, HTML/CSS, Spring Boot, Flask
- **Data Analysis & Visualization** pandas, NumPy, Matplotlib, seaborn, Jupyter Notebook
- **NLP/ML** BERTopic, Word2Vec, S-WEAT, Topic Modeling, Text Mining
- **Languages** Chinese (native), English (fluent)